KAIZEN MATHS

Lesson starter

Linear Programming

Context Problems

Name _____

SA1

Sketch the feasible region using the constraints.

A workshop makes tables (x) and chairs (y). The constraints show weekly production limits. Profit is £4 per table and £7 per chair. Find the production plan that maximises profit.

$0 \leq x \leq 11$

$0 \leq y \leq 8$

$3x + 2y \leq 43$

Maximise $P = 4x + 7y$

ALG-AFEA-L3 · SA

Show your working

KAIZEN MATHS

Lesson starter

Linear Programming

Context Problems

Name _____

SB1

Sketch the feasible region using the constraints.

A diet plan uses food A (x portions) and food B (y portions). The constraints show minimum nutritional requirements and maximum portions available. Food A costs £9 and food B costs £8. Find the least cost.

$0 \leq x \leq 8$

$0 \leq y \leq 10$

$x + 4y \geq 4$

Minimise $C = 9x + 8y$

ALG-AFEA-L3 · SB

Show your working

KAIZEN MATHS

Lesson starter

Linear Programming

Context Problems

Name _____

SC1

Sketch the feasible region using the constraints.

The variables x and y must be whole numbers. Maximise P = 4x + 7y subject to the constraints.

$0 \leq x \leq 9$

$0 \leq y \leq 10$

$4x + 4y \leq 60$

Whole-number solution required

KAIZEN MATHS

Lesson starter

Linear Programming

Context Problems

Name _____

SD1

Sketch the feasible region using the constraints.

A workshop makes tables (x) and chairs (y). The constraints show weekly production limits. Profit is £8 per table and £5 per chair. Find the production plan that maximises profit.

$0 \leq x \leq 6$

$0 \leq y \leq 9$

$2x + y \leq 17$

Maximise P = 8x + 5y

KAIZEN MATHS

Exit ticket

Linear Programming

Context Problems

Name _____

EA1

Sketch the feasible region using the constraints.

A diet plan uses food A (x portions) and food B (y portions). The constraints show minimum nutritional requirements and maximum portions available. Food A costs £3 and food B costs £4. Find the least cost.

$0 \leq x \leq 7$

$0 \leq y \leq 11$

$3x + 3y \geq 9$

Minimise C = 3x + 4y

ALG-AFEA-L3 · EA

Show your working

KAIZEN MATHS

Exit ticket

Linear Programming

Context Problems

Name _____

EB1

Sketch the feasible region using the constraints.

The variables x and y must be whole numbers. Maximise P = 7x + 9y subject to the constraints.

$0 \leq x \leq 6$

$0 \leq y \leq 8$

$x + y \leq 12$

Whole-number solution required

ALG-AFEA-L3 · EB

Show your working

KAIZEN MATHS

Exit ticket

Linear Programming

Context Problems

Name _____

EC1

Sketch the feasible region using the constraints.

A workshop makes tables (x) and chairs (y). The constraints show weekly production limits. Profit is £4 per table and £7 per chair. Find the production plan that maximises profit.

$0 \leq x \leq 8$

$0 \leq y \leq 9$

$3x + 2y \leq 36$

Maximise P = 4x + 7y

ALG-AFEA-L3 · EC

Show your working

KAIZEN MATHS

Exit ticket

Linear Programming

Context Problems

Name _____

EA1

Sketch the feasible region using the constraints.

A diet plan uses food A (x portions) and food B (y portions). The constraints show minimum nutritional requirements and maximum portions available. Food A costs £3 and food B costs £4. Find the least cost.

$0 \leq x \leq 7$

$0 \leq y \leq 11$

$3x + 3y \geq 9$

Minimise C = 3x + 4y

ALG-AFEA-L3 · EA

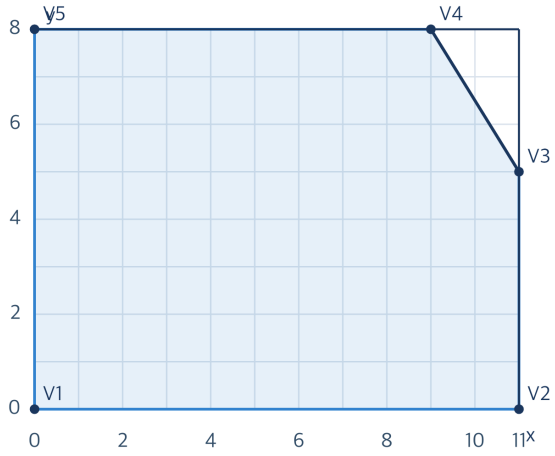
Show your working

Teacher answers & working

Linear Programming · Context Problems

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SA1



A workshop makes tables (x) and chairs (y). The constraints show weekly production limits. Profit is £4 per table and £7 per chair. Find the production plan that maximises profit.

$0 \leq x \leq 11$

$0 \leq y \leq 8$

$3x + 2y \leq 43$

Maximise P = 4x + 7y

Answer:

Make 9 tables and 8 chairs for maximum profit £92.

Working:

Let x be the number of tables and y be the number of chairs.

The feasible region shows all production plans that satisfy the limits.

Vertex	P	Value
(0, 0)	4(0) + 7(0)	0
(11, 0)	4(11) + 7(0)	44
(11, 5)	4(11) + 7(5)	79
(9, 8)	4(9) + 7(8)	92
(0, 8)	4(0) + 7(8)	56

The maximum profit is £92 at (9, 8).

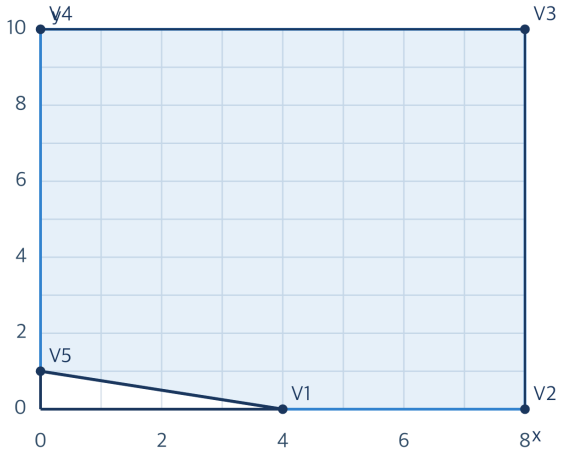
Interpretation: make 9 tables and 8 chairs.

Teacher answers & working

Linear Programming · Context Problems

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SB1



A diet plan uses food A (x portions) and food B (y portions). The constraints show minimum nutritional requirements and maximum portions available. Food A costs £9 and food B costs £8. Find the least cost.

$0 \leq x \leq 8$

$0 \leq y \leq 10$

$x + 4y \geq 4$

Minimise C = 9x + 8y

Answer:

Use 0 portions of A and 1 portions of B for minimum cost £8.

Working:

Let x be portions of food A and y be portions of food B.

Because this is a minimum-requirement problem, the feasible region is on the side satisfying the ≥ constraint.

Vertex	C	Value
(4, 0)	9(4) + 8(0)	36
(8, 0)	9(8) + 8(0)	72
(8, 10)	9(8) + 8(10)	152
(0, 10)	9(0) + 8(10)	80
(0, 1)	9(0) + 8(1)	8

The minimum cost is £8 at (0, 1).

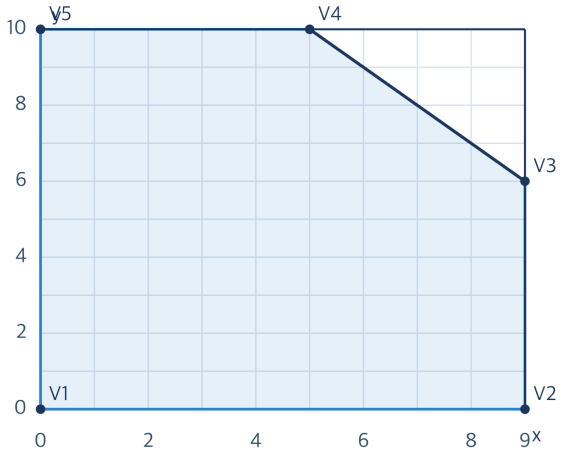
Interpretation: use 0 portions of A and 1 portions of B.

Teacher answers & working

Linear Programming · Context Problems

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SC1



The variables x and y must be whole numbers. Maximise P = 4x + 7y subject to the constraints.

$0 \leq x \leq 9$

$0 \leq y \leq 10$

$4x + 4y \leq 60$

Whole-number solution required

Answer:

Best integer solution is (5, 10) with P = 90.

Working:

The graph gives the continuous feasible region, but the question asks for whole-number values.

Check feasible lattice points near the best continuous vertex, or systematically test all feasible whole-number points.

Number of feasible whole-number points checked: 100.

Best integer point: (5, 10).

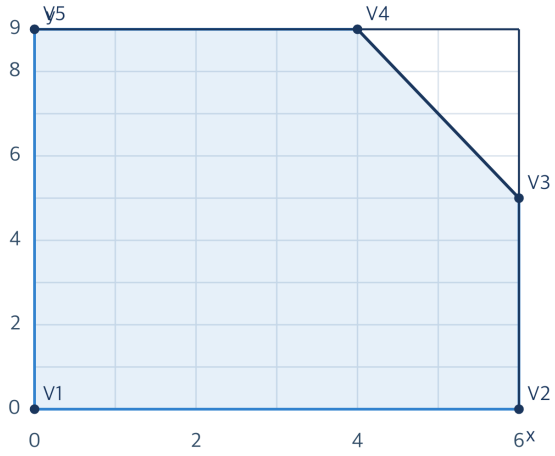
$P = 4(5) + 7(10) = 90.$

Teacher answers & working

Linear Programming · Context Problems

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SD1



A workshop makes tables (x) and chairs (y). The constraints show weekly production limits. Profit is £8 per table and £5 per chair. Find the production plan that maximises profit.

$0 \leq x \leq 6$

$0 \leq y \leq 9$

$2x + y \leq 17$

Maximise P = 8x + 5y

Answer:

Make 4 tables and 9 chairs for maximum profit £77.

Working:

Let x be the number of tables and y be the number of chairs.

The feasible region shows all production plans that satisfy the limits.

Vertex	P	Value
(0, 0)	8(0) + 5(0)	0
(6, 0)	8(6) + 5(0)	48
(6, 5)	8(6) + 5(5)	73
(4, 9)	8(4) + 5(9)	77
(0, 9)	8(0) + 5(9)	45

The maximum profit is £77 at (4, 9).

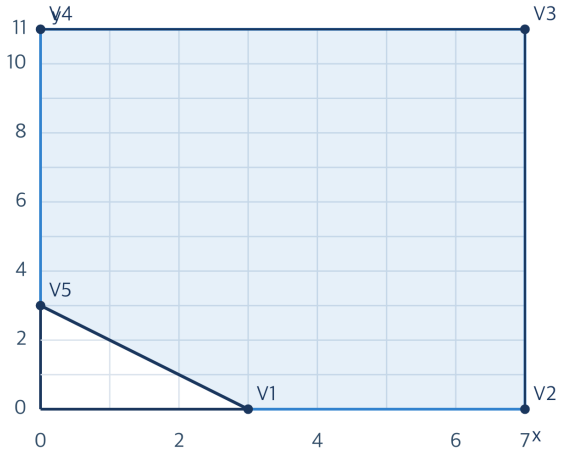
Interpretation: make 4 tables and 9 chairs.

Teacher answers & working

Linear Programming · Context Problems

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EA1



A diet plan uses food A (x portions) and food B (y portions). The constraints show minimum nutritional requirements and maximum portions available. Food A costs £3 and food B costs £4. Find the least cost.

$0 \leq x \leq 7$

$0 \leq y \leq 11$

$3x + 3y \geq 9$

Minimise C = 3x + 4y

Answer:

Use 3 portions of A and 0 portions of B for minimum cost £9.

Working:

Let x be portions of food A and y be portions of food B.

Because this is a minimum-requirement problem, the feasible region is on the side satisfying the ≥ constraint.

Vertex	C	Value
(3, 0)	3(3) + 4(0)	9
(7, 0)	3(7) + 4(0)	21
(7, 11)	3(7) + 4(11)	65
(0, 11)	3(0) + 4(11)	44
(0, 3)	3(0) + 4(3)	12

The minimum cost is £9 at (3, 0).

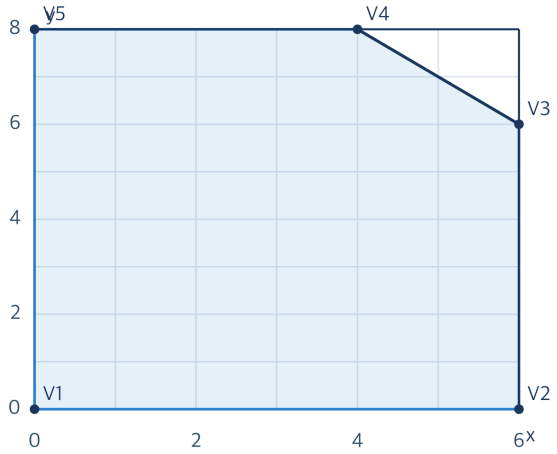
Interpretation: use 3 portions of A and 0 portions of B.

Teacher answers & working

Linear Programming · Context Problems

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EB1



The variables x and y must be whole numbers. Maximise P = 7x + 9y subject to the constraints.

$0 \leq x \leq 6$

$0 \leq y \leq 8$

$x + y \leq 12$

Whole-number solution required

Answer:

Best integer solution is (4, 8) with P = 100.

Working:

The graph gives the continuous feasible region, but the question asks for whole-number values.

Check feasible lattice points near the best continuous vertex, or systematically test all feasible whole-number points.

Number of feasible whole-number points checked: 60.

Best integer point: (4, 8).

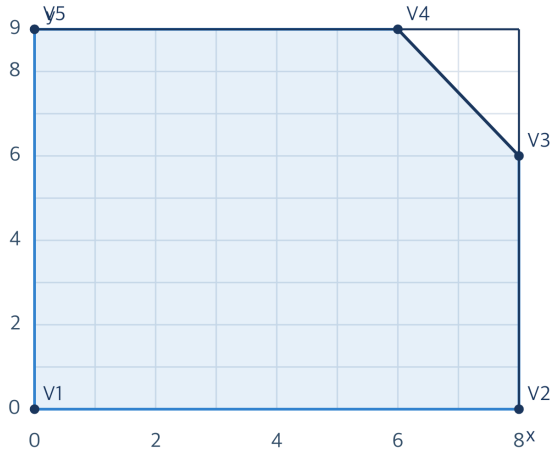
$P = 7(4) + 9(8) = 100.$

Teacher answers & working

Linear Programming · Context Problems

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EC1



A workshop makes tables (x) and chairs (y). The constraints show weekly production limits. Profit is £4 per table and £7 per chair. Find the production plan that maximises profit.

$0 \leq x \leq 8$

$0 \leq y \leq 9$

$3x + 2y \leq 36$

Maximise P = 4x + 7y

Answer:

Make 6 tables and 9 chairs for maximum profit £87.

Working:

Let x be the number of tables and y be the number of chairs.

The feasible region shows all production plans that satisfy the limits.

Vertex	P	Value
(0, 0)	4(0) + 7(0)	0
(8, 0)	4(8) + 7(0)	32
(8, 6)	4(8) + 7(6)	74
(6, 9)	4(6) + 7(9)	87
(0, 9)	4(0) + 7(9)	63

The maximum profit is £87 at (6, 9).

Interpretation: make 6 tables and 9 chairs.